

PROJECT TITLE :

Analysis of Adidas Sales Performance

Objectives

- To analyze Adidas sales data to understand overall business performance
- To identify top-performing products, regions, and cities
- To study the relationship between sales, profit, and pricing
- To analyze customer purchasing patterns and sales methods
- To derive meaningful insights for better decision-making

Introduction

Adidas is a globally recognized brand known for its sportswear and footwear products. Analyzing sales data helps in understanding market trends, customer behavior, and business performance.

This project focuses on exploring Adidas sales data using data analysis techniques and visualizations to uncover patterns, trends, and insights that can support strategic decisions.

Data Source

The dataset used in this project is the **Adidas Sales Dataset**, which is publicly available and commonly used for data analysis practice.

- Source: Kaggle
- Format: Excel file
- Type: Structured dataset

The dataset contains real-world-like sales transaction data, making it suitable for business analysis and visualization.

Project Structure

The project is organized into the following steps:

1. **Data Collection**
 - Obtaining the Adidas sales dataset
2. **Data Cleaning & Preprocessing**
 - Handling missing values
 - Converting data types (e.g., date format)
 - Creating new features (e.g., Month)
3. **Exploratory Data Analysis (EDA)**
 - Understanding data distribution
 - Identifying patterns and relationships
4. **Data Visualization**
 - Creating charts (bar, line, pie, scatter)
 - Comparing different variables
5. **Insight Generation**
 - Interpreting graphs
 - Identifying trends and patterns
6. **Conclusion & Recommendations**
 - Summarizing findings
 - Suggesting business improvements

Dataset Description

The dataset contains the following key fields:

- Invoice Date
- Product
- Region, State, City
- Sales Method
- Units Sold

- Price per Unit
- Total Sales
- Operating Profit
- Operating Margin

Tools & Technologies Used

- Python
- Pandas
- Matplotlib
- Colab

Data Preprocessing

- Converted *Invoice Date* into datetime format
- Checked for missing or inconsistent values
- Created new column: **Month** for trend analysis
- Grouped data for analysis

Data Analysis & Visualization

Total Sales Over Time

Sales by Region

Sales by Product

Sales vs Operating Profit

Sales Method Contribution

Monthly Sales Trend

Operating Margin by Product

Price vs Units Sold

Sales by State

Top 10 Cities by Sales

Units Sold by Product

Key Insights

- Sales performance varies across products and regions
- Certain products dominate revenue generation
- Profitability is not always proportional to sales
- Seasonal trends affect monthly sales
- Pricing impacts demand significantly

Findings

- Strong sales concentration in specific regions
- Uneven contribution of sales methods
- High variation in product performance
- Demand influenced by pricing

Recommendations

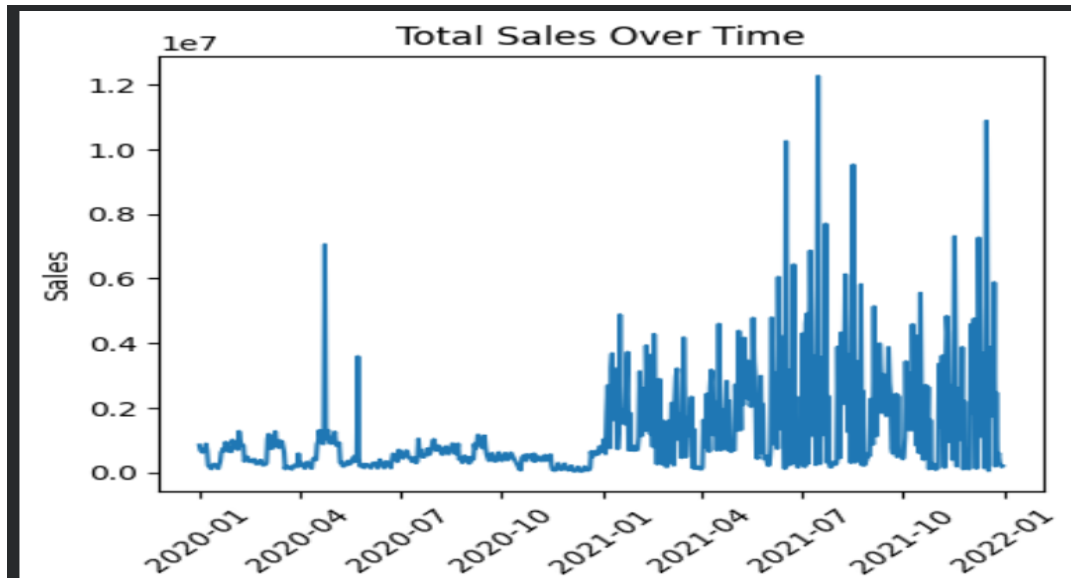
- Focus on high-performing products
- Improve strategies in low-performing regions
- Optimize pricing for better sales volume
- Invest in effective sales channels
- Reduce costs for low-margin products

Project Report: Analytical Questions with Explanation

1. Total Sales Over Time (Line Chart)

Question: How does total sales change over time?

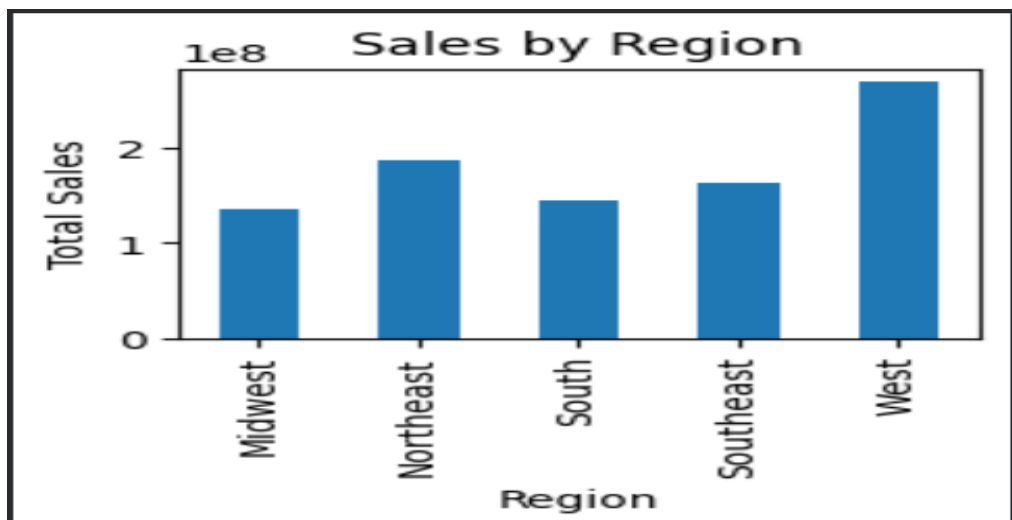
Explanation: This helps identify overall business performance trends such as growth, decline, or fluctuations, which are important for forecasting and strategic planning.



2. Sales by Region (Bar Chart)

Question: Which region generates the highest total sales?

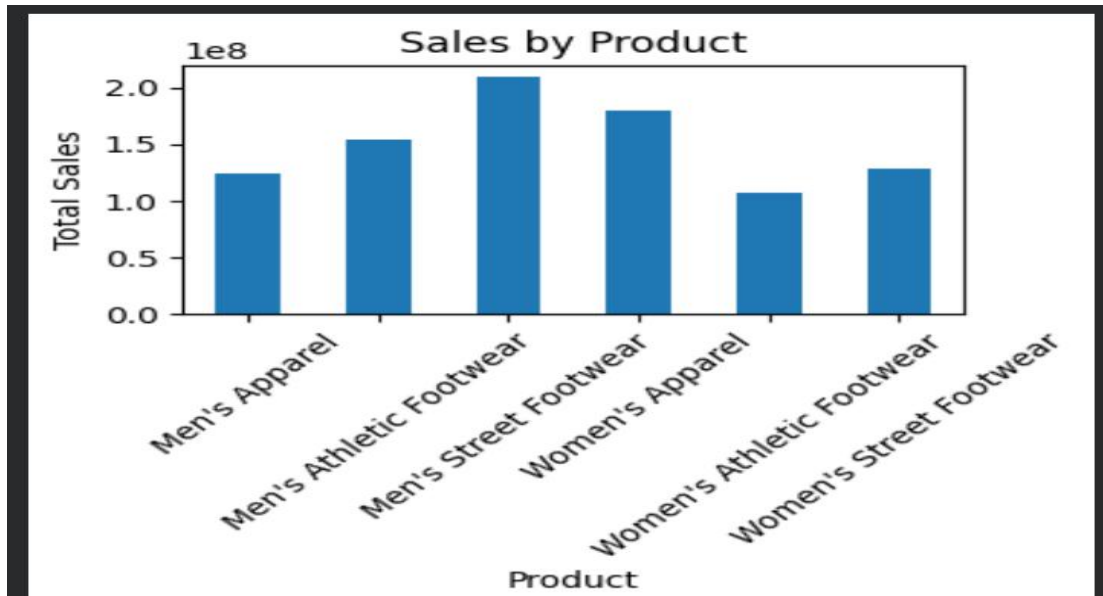
Explanation: This identifies the strongest geographic markets and helps businesses focus resources on high-performing regions.



3. Sales by Product (Bar Chart)

Question: Which product contributes the most to total sales?

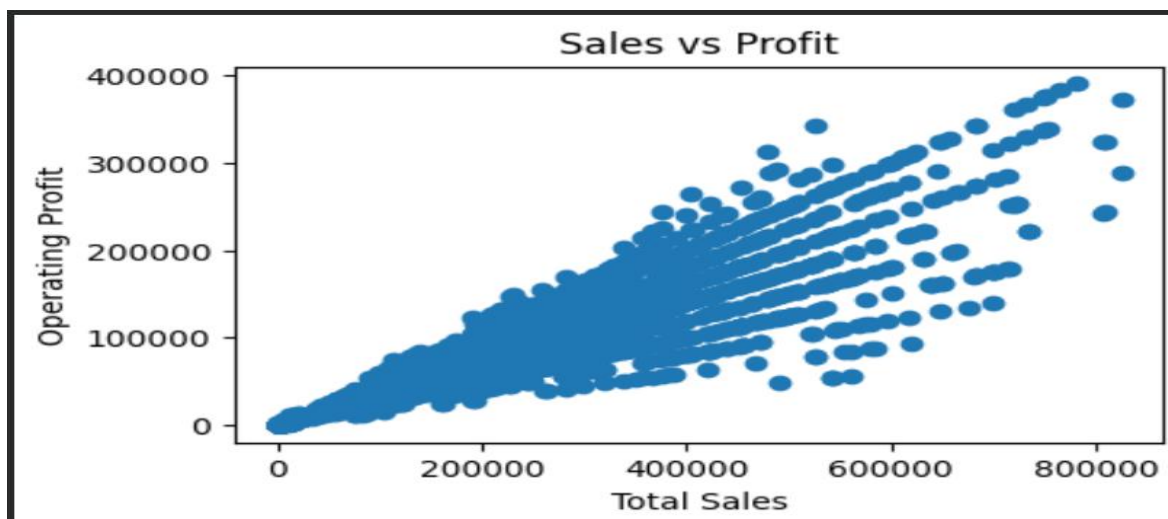
Explanation: This reveals top-performing products and helps prioritize inventory, marketing, and production decisions.



4. Sales vs Operating Profit (Scatter Plot)

Question: Does higher sales always result in higher operating profit?

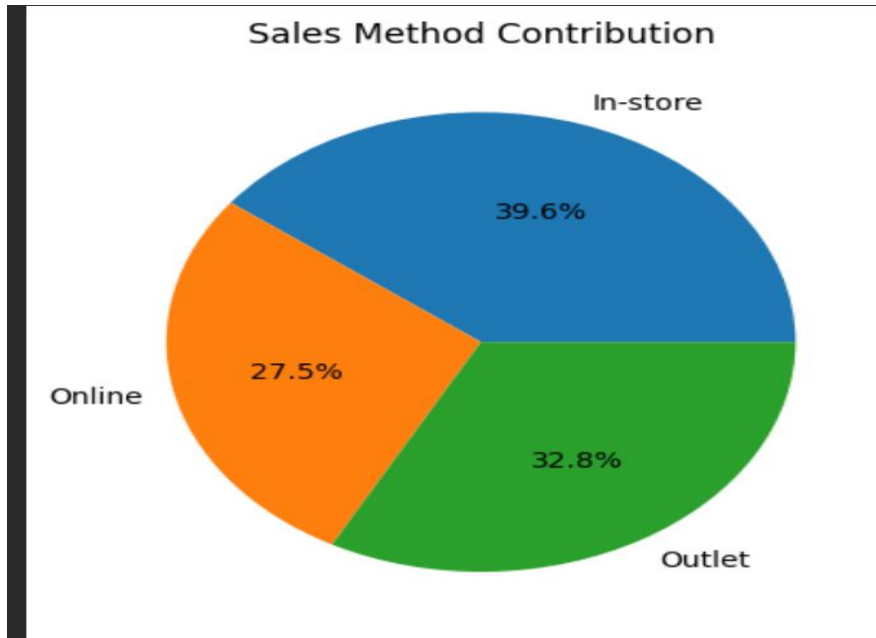
Explanation: This shows whether increased revenue leads to profitability or if costs are reducing overall profit.



🤖 5. Sales Method Contribution (Pie Chart)

Question: Which sales method contributes the largest share of total sales?

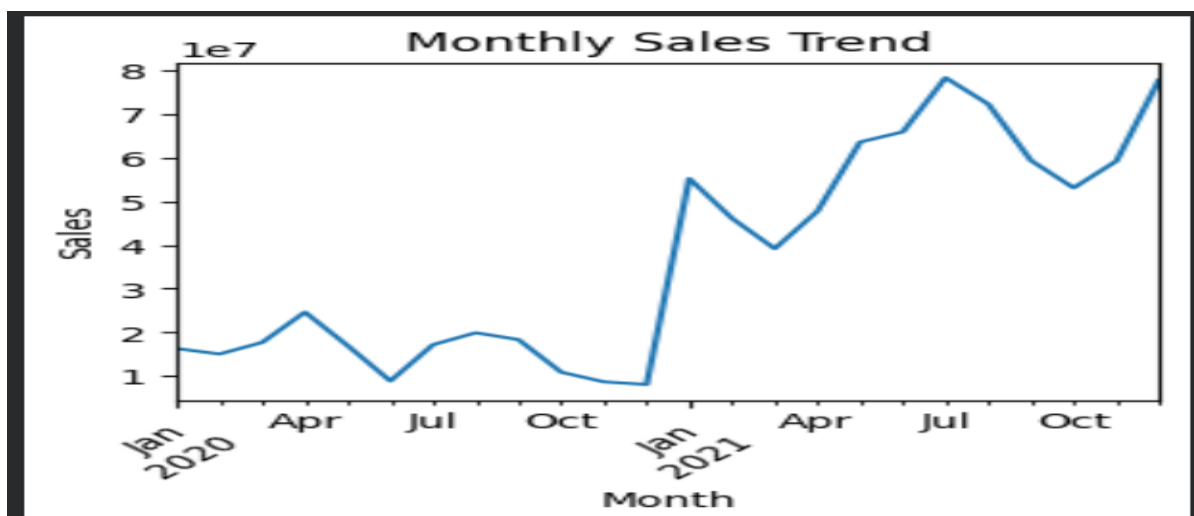
Explanation: This helps identify the most effective sales channel and supports better marketing and distribution strategies.



📅 6. Monthly Sales Trend (Line Chart)

Question: What is the monthly trend of total sales?

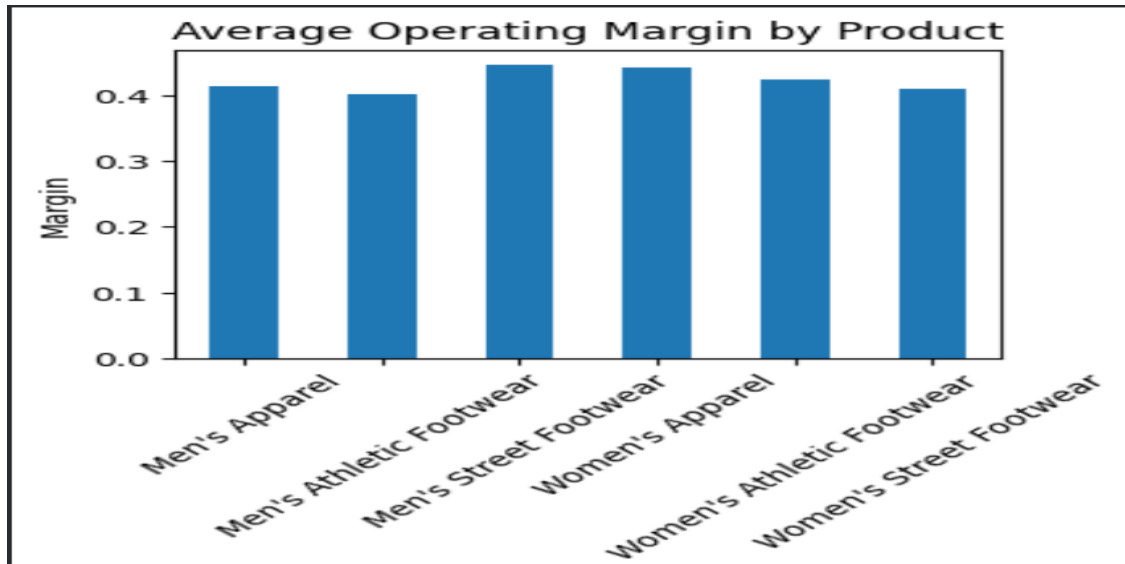
Explanation: This helps detect seasonal patterns and monthly fluctuations, useful for demand forecasting and planning.



7. Operating Margin by Product (Bar Chart)

Question: Which product has the highest average operating margin?

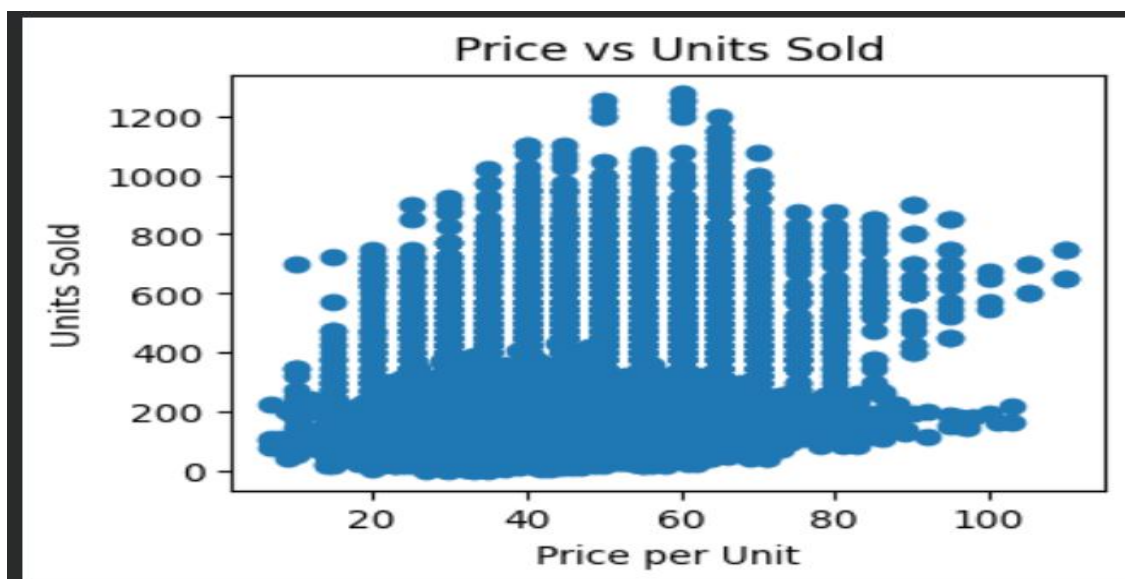
Explanation: This identifies the most profitable products and highlights efficiency in cost management.



8. Price vs Units Sold (Scatter Plot)

Question: How does price affect the number of units sold?

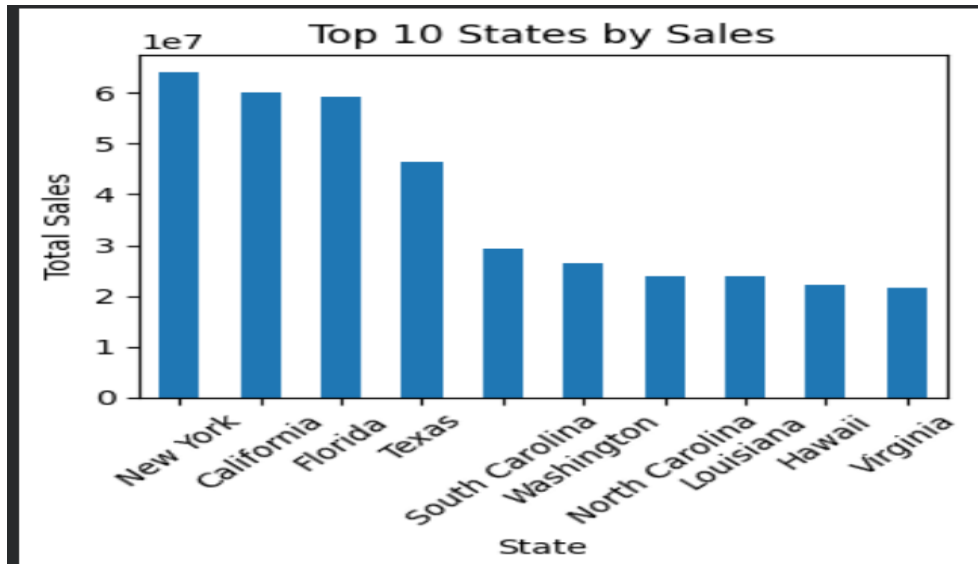
Explanation: This shows the relationship between pricing and demand, helping in setting optimal pricing strategies.



9. Sales by State (Bar Chart)

Question: Which state contributes the most to total sales?

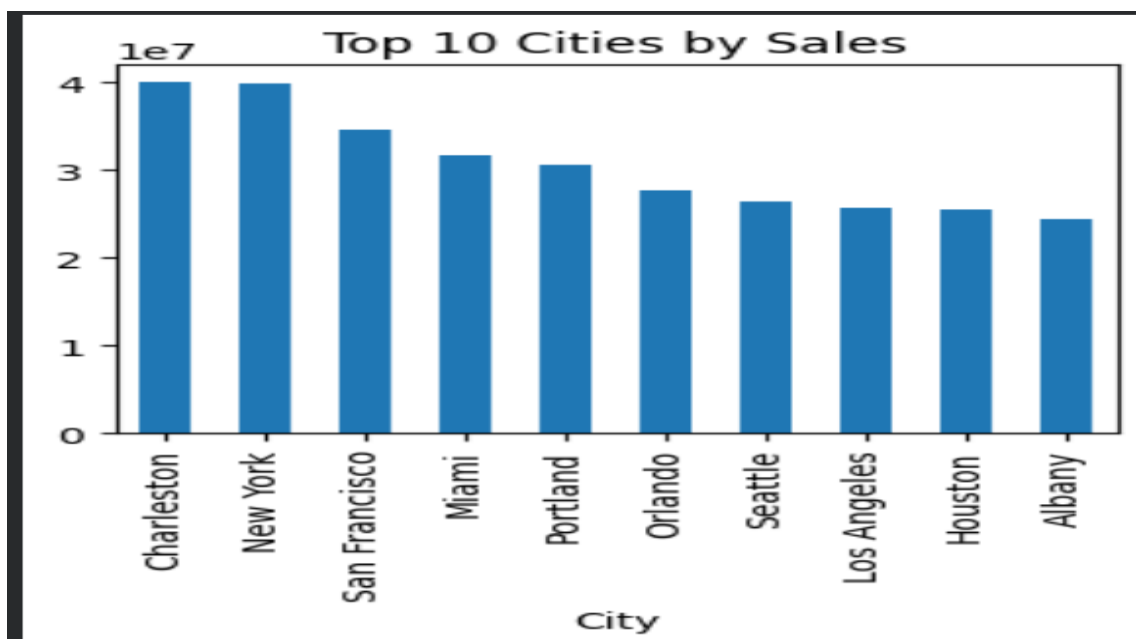
Explanation: This highlights regional performance and helps identify areas for expansion or improvement.



10. Top 10 Cities by Sales (Bar Chart)

Question: Which cities are the top contributors to total sales?

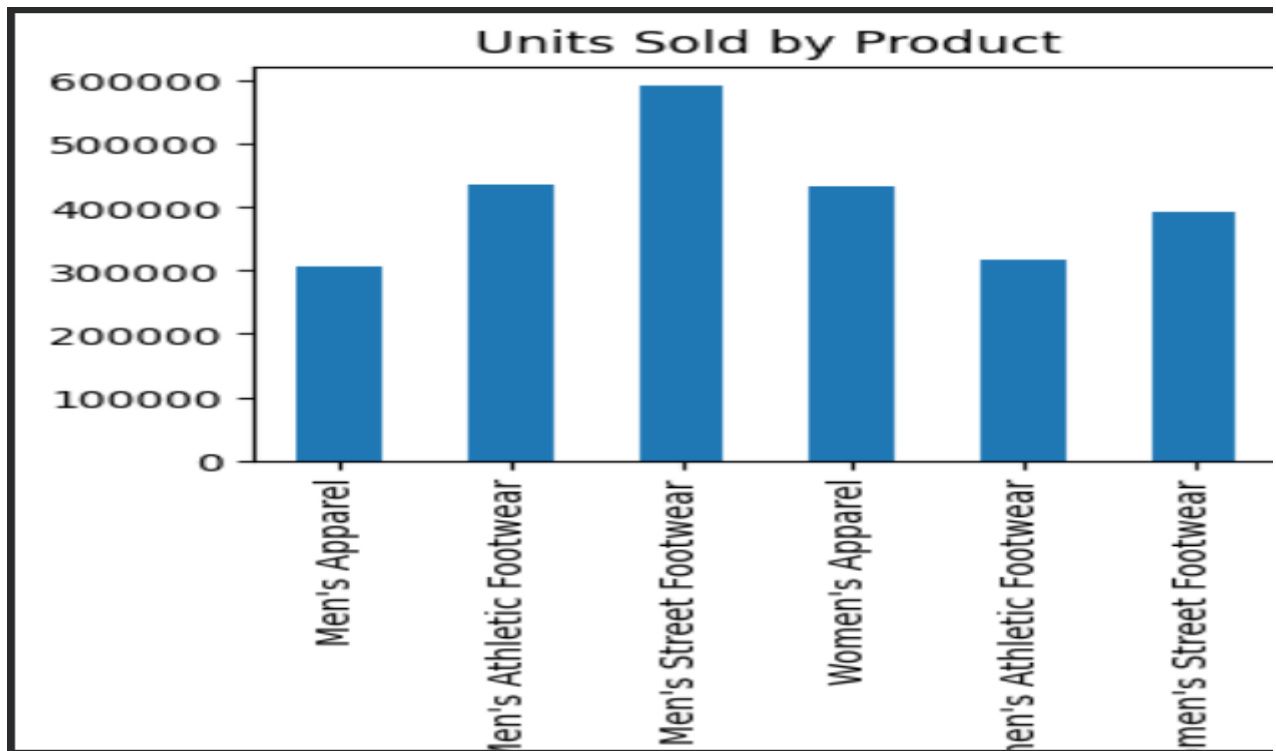
Explanation: This identifies key urban markets driving revenue and supports targeted business strategies.



📦 11. Units Sold by Product (Bar Chart)

Question: Which product has the highest number of units sold?

Explanation: This shows product demand and helps compare sales volume with revenue and profitability.



Conclusion

This project analyzed Adidas sales data to uncover key trends in sales, profitability, and customer behavior.

The findings highlight important business insights that can support strategic decision-making. Data analysis plays a crucial role in improving efficiency and maximizing business performance.